

Coordinated regulation of gene expression in *Plasmodium* female gametocytes by two transcription factors

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Abstract

Gametocytes play key roles in the *Plasmodium* lifecycle. They are essential for sexual reproduction as precursors of the gametes. They also play an essential role in parasite transmission to mosquitoes. Elucidation of the gene regulation at this stage is essential for understanding these two processes at the molecular level and for developing new strategies to break the parasite life cycle. We identified a novel *Plasmodium* transcription factor (TF), designated as a partner of AP2-FG or PFG. In this paper, we report that this TF regulates the gene expression in female gametocytes in concert with another female-specific TF AP2-FG. Upon the disruption of *PFG*, majority of female-specific genes were significantly downregulated, and female gametocyte lost the ability to produce ookinetes. ChIP-seq analysis showed that it was located in the same position as AP2-FG, indicating that these two TFs form a complex. ChIP-seq analysis of PFG in *AP2-FG*-disrupted parasites and ChIP-seq analysis of AP2-FG in *PFG*-disrupted parasites demonstrated that PFG mediates the binding of AP2-FG to the ten-base motif and that AP2-FG binds another motif, GCTCA, in the absence of PFG. In promoter assays, this five-base motif was identified as another female-specific cis-acting element. Genes under the control of the two forms of AP2-FG, with or without PFG, partly overlapped; however, each form had target preferences. These results suggested that combinations of these two forms generate various expression patterns among the extensive genes expressed in female gametocytes.

eLife assessment

This study offers **important** insights into the transcriptional regulatory networks driving female gametocyte maturation in rodent malaria parasites. The work is based on **solid** methodology and shows how two female-specific transcription factors, AP2-FG and PFG (aka Fd2), co-operate to up-regulate the expression of genes required for development after fertilization occurs in the mosquito midgut. This study will be of interest to scientists working on sexual differentiation and gene regulation in *Plasmodium* and other apicomplexan parasites.

Introduction

Plasmodium gametocytes are erythrocytic sexual stages essential for parasite transmission to mosquito vectors. Within the blood meal of a mosquito, female and male gametocytes egress host erythrocytes and transform into gametes for fertilization. Zygotes undergo meiotic nuclear division and develop into motile ookinetes to escape from the blood meal to the mosquito midgut epithelium. The elucidation of gene regulation during this lifecycle stage is important for developing antimalarial strategies that block transmission and prevent the spread of the disease.

Gametocytes develop from a subset of asexual erythrocytic parasites that are committed to sexual reproduction. The transition from the asexual erythrocytic stage to gametocytes occurs in late trophozoites; a subpopulation of trophozoites differentiates into gametocytes for sexual reproduction, and the rest develop into schizonts to continue asexual proliferation. Early gametocytes display no sex-specific features, but after a period of time, which is different among *Plasmodium* species, they start differentiating into each sex and finally become mature male and female gametocytes that are ready to transform into gametes. In *P. berghei*, early gametocytes manifest morphologies distinguishable from asexual stages at 18 hpi (hours post-infection of the new red blood cell) and manifest sex-specific features at 20 hpi. They then complete a sex-specific maturation process, during which a sex-specific gene expression repertoire is established, at approximately 24–26 hpi (1). Sex-specific proteomic and transcriptomic analyses in *P. berghei* have revealed a number of genes specifically expressed in each sex (2)(3)(4). In females, genes necessary for fertilization and protein synthesis are expressed, and the number of transcripts is stored in an untranslated state in the cytoplasm to prepare for zygote development (5)(6). In males, genes for flagella-based motility and genome replication are expressed, as well as genes for fertilization.

Recently, it has become clear that transcription factors (TFs) belonging to the AP2 family play a central role in sexual development. AP2-G is essential (7). Disruption of this gene causes the parasite to lose its ability to generate gametocytes. In *P. berghei*, its expression starts from the late trophozoite at 14–16 hpi and peaks at 18 hpi when sexual stage-specific features first appear (8). *P. berghei* AP2-G induces hundreds of genes that contain several TFs important for the development of gametocytes, suggesting that induction of these TFs by AP2-G generates a driving force for gametocytogenesis (8).

AP2-FG is a member of the *Plasmodium* AP2 family of TFs and is a target genes of AP2-G (8)(9). AP2-FG is a female-specific TF and plays an essential role in the development of female gametocytes. In *P. berghei*, AP2-FG is expressed from early gametocytes to mature females, and by disruption of *AP2-FG*, the development of the female gametocytes is impaired, showing immature morphologies and resulting in complete loss of capability to mediate parasite transmission to mosquitoes. This TF binds to a ten-base female-specific *cis*-acting element and regulates variety of genes include those for fertilization, meiosis, and the development of ookinetes. This broad repertoire of target genes and their essential role in female development suggest that this TF is a master regulator of female development. However, *AP2-FG*-disrupted parasites can produce ookinetes with decreased numbers and abnormal morphologies, suggesting that additional mechanisms for activating female-specific genes are still functional upon the disruption of *AP2-FG*.

We explored novel TFs for gametocyte development in the target genes of AP2-G and identified a novel female gametocyte-specific TF. In this paper, we report that this TF, designated as PFG, regulates the gene expression in female gametocytes in concert with AP2-FG.

Results

***PFG* is a target gene of AP2-G and is expressed in female gametocytes**

We screened for novel sequence-specific TFs among target genes that were functionally unannotated in PlasmoDB (<https://plasmodb.org/plasmo/app/>) using highly conserved amino acid sequences among *Plasmodium* species as a criterion. We hypothesized that the amino acid sequences of DNA-binding domains of sequence-specific TFs would have been difficult to change during evolution because even a small change in the sequence-specificity of TFs could cause catastrophic effects by global changes in gene expression. Through this screening, a gene encoding a 2709 amino acid protein with two regions highly conserved among *Plasmodium* was identified (PBANKA0902300, designated as a *partner of AP2-FG* (*PFG*; **Fig. 1A**)). This gene is one of the *P. berghei* genes that were previously identified as genes involved in female gametocyte development (named *FD2*), based on mass screening combined with single cell RNA-seq (10). The two conserved regions comprised 108- and 139-amino acids, respectively, and each region showed 94% and 91% conservation between *P. berghei* and *P. falciparum* (**Fig. 1B**). The Blastp search (<https://blast.ncbi.nlm.nih.gov/Blast.cgi>) using these regions revealed that genes homologous to *PFG* exist broadly among apicomplexan parasites and also in alveolates closely related to Apicomplexa, such as *Vitrella brassicaformis* (**Fig. 1C**), whereas a large inter-region between these two regions present in all *Plasmodium* homologs was not observed in these proteins (**Fig. 1A**). Based on domain prediction using the SMART database (<http://smart.embl-heidelberg.de/>), no functionally characterized domains were identified to be significantly similar to these two regions. To investigate the expression of *PFG*, we generated parasites expressing GFP-fused *PFG* (*PFG::GFP* parasites; **Fig. S1A**). Fluorescence analysis showed that this gene is expressed only in female gametocytes and is localized in the nucleus (**Fig. 1D**), suggesting its involvement in the transcriptional regulation of female gametocytes. According to sex-specific transcriptome data (3), the *PFG* is specifically transcribed in female gametocytes. We examined the target genes of the female-specific transcriptional activator AP2-FG and found that it had been predicted as a target of AP2-FG and harbored AP2-FG peaks with binding motifs in the upstream region (**Fig. 1E**). These results suggest that *PFG* was activated in two steps, that is, by AP2-G and then by AP2-FG, during female development. Temporal profiling of the expression of *PFG* using *PFG::GFP* parasites showed that the expression became visible from 20 hpi (hours post-erythrocyte infection) (**Fig. 1F**). The timing of the expression was approximately four hours later than that of AP2-FG, which started at 16 hpi (9).

***PFG* is essential for female development**

We generated *PFG*-disrupted (*PFG*⁻) parasites to assess the role of *PFG* (**Fig. S1B**). The asexual stages of *PFG*⁻ parasites proliferated as wild-types (**Fig. 2A**), and morphologically mature male and female gametocytes were observed on Giemsa-stained blood smears (**Fig. 2B**). However, in mosquito transmission experiments, no oocysts were observed on the mosquito midgut wall 14 days after the infective blood meal in two independent clones (**Fig. 2C**). In ookinete cultures, only approximately 20% of the female gametocytes were converted into zygotes, and no banana-shaped or retort-form ookinetes were generated (**Fig. 2D**). Considering the female-specific expression of *PFG*, these results suggest that disruption of *PFG* significantly impaired the development of female gametocytes. In the cross-fertilization experiment, ookinetes were formed when crossing with normal females (*P48/45*⁻) but not when crossing with normal males (*P47*⁻), demonstrating that the abnormal phenotype observed in *PFG*⁻ was indeed derived from female gametocytes (**Fig. 2E**).

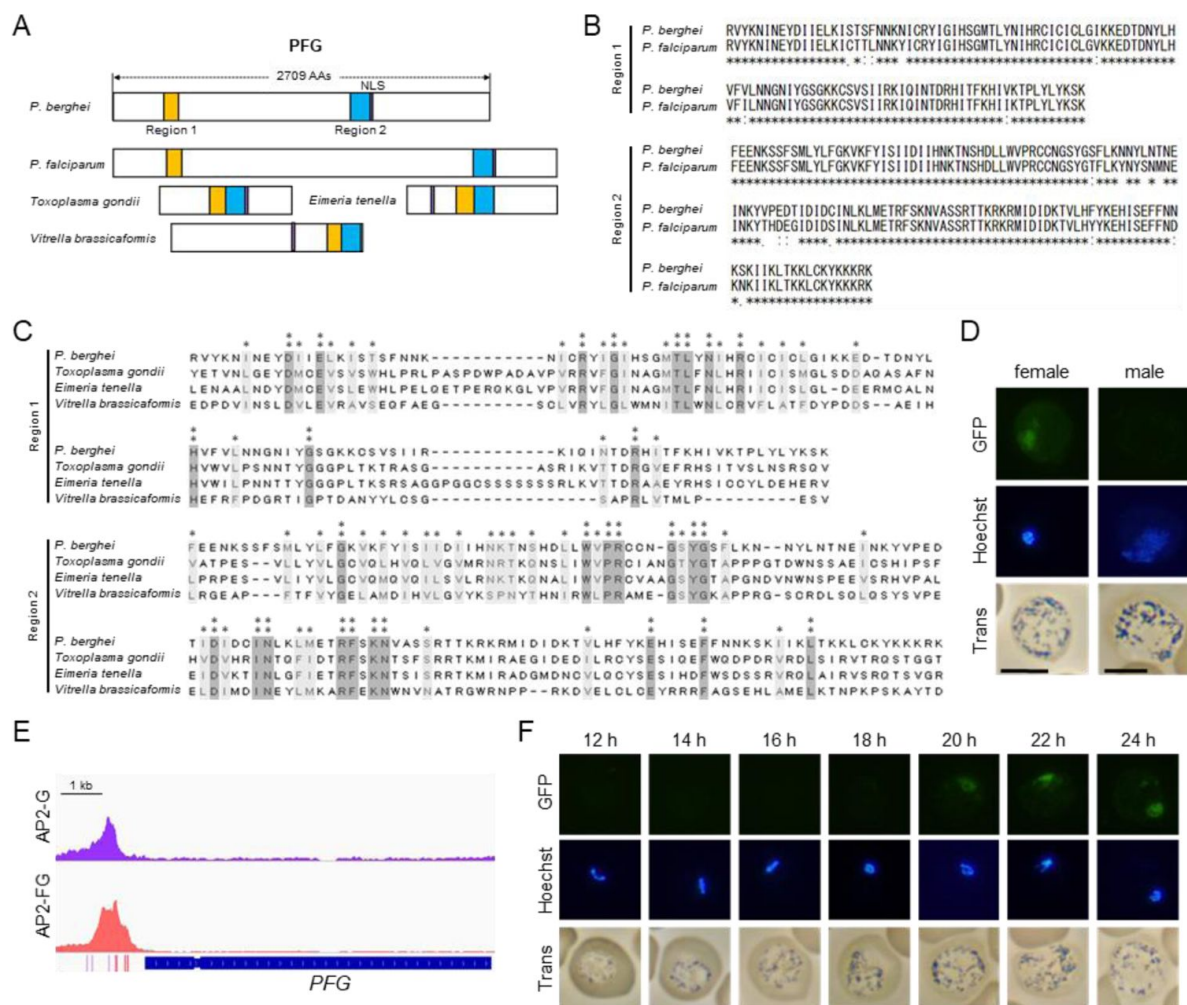


Figure 1.

PFG is a target gene of AP2-G and is expressed in female gametocytes

A. Schematic diagram of PFG from *P. berghei* and its homologs in apicomplexan parasites. Regions homologous to Regions 1 and 2, which are highly conserved among *Plasmodium* species, are shown as yellow and blue rectangles, respectively. Nuclear localization signals were predicted using the cNLS mapper (http://nls-10mapper.iab.keio.ac.jp/cgi-bin/NLS_Mapper_form.cgi). The gene IDs of *P. berghei* PFG, *P. falciparum* PFG, and their homologs in *Toxoplasma gondii*, *Eimeria tenella* and *Vitrella brassicaformis* are PBANKA_0902300, PF3D7_1146800, TGGT1_239670, ETH2_1252400, and Vbra_10234, respectively. B. Alignment of amino acid sequences of Region 1 and Region 2 from *P. berghei* and *P. falciparum* PFGs was performed using the ClustalW program (<https://www.genome.jp/tools-bin/clustalw>). C. The amino acid sequences of Regions 1 and 2 from *P. berghei* PFG and its homologs from other apicomplexan parasites in (A) were aligned using the ClustalW program in MEGA X. The positions at which all these sequences have identical amino acids are indicated by two asterisks, and positions with amino acid residues possessing the same properties are indicated by one asterisk. D. Expression of PFG in mature male and female gametocytes of PFG::GFP parasites. Nuclei were stained with Hoechst 33342. Scale bars: 5 μ m. E. Integrative Genomics Viewer (IGV) images from the ChIP-seq data of AP2-G and AP2-FG in the upstream region of PFG. Data were obtained from previous papers (8)(9). The purple and red bars indicate the binding motifs of AP2-G and AP2-FG in the peak region. F. Time-course observations of PFG expression during gametocyte development using PFG::GFP parasites. Parasites were observed every two hours from 12 hpi onwards. Nuclei were stained with Hoechst 33342.

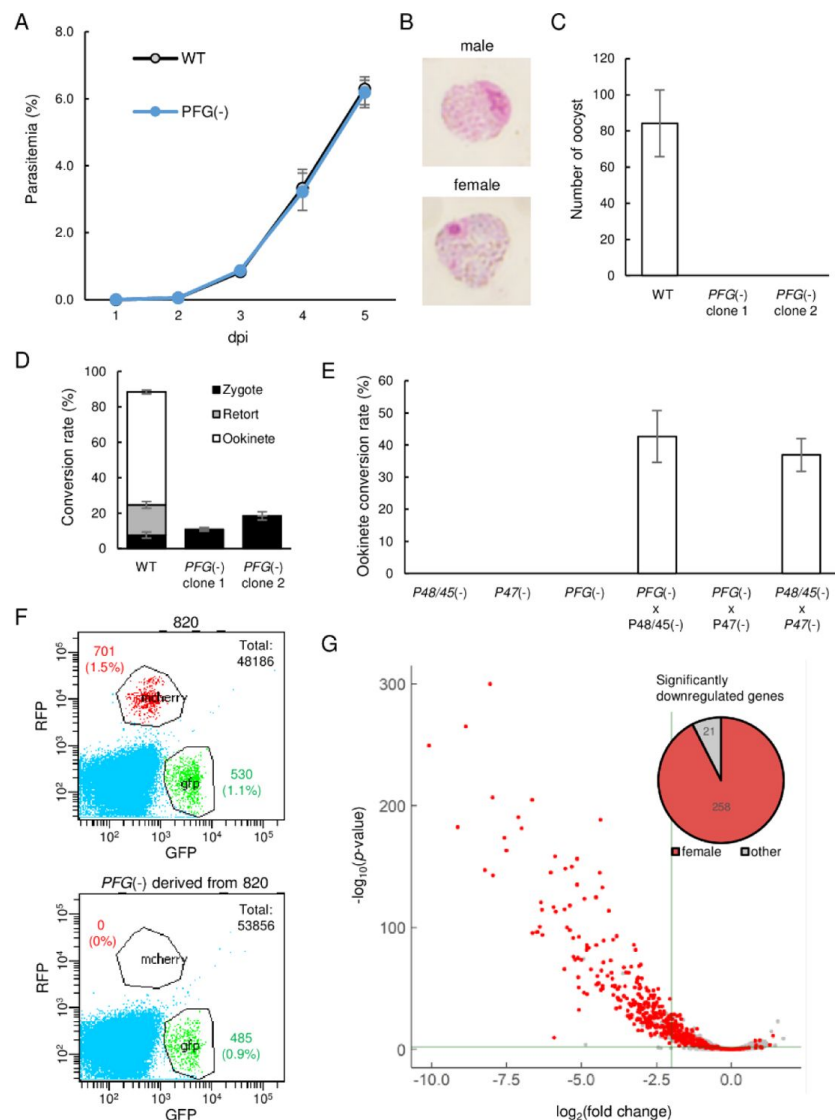


Figure 2.

Development of female gametocytes was impaired in *PFG*(-) parasites.

A. Proliferation of blood-stage parasites in wild-type and *PFG*(-) parasites. Parasitemia was calculated by counting infected red blood cells using Giemsa staining of blood smears. B. Giemsa-stained images of mature male and female gametocytes in *PFG*(-) parasites. C. Number of oocysts in the mosquito midgut 14 days after infective blood meal of wild-type and *PFG*(-) parasites. D. Ookinete cultures of wild type and *PFG*(-) parasites. Ratios of zygotes, retort-form, and banana-shaped ookinetes to all female-derived cells are shown in black, grey, and white, respectively. E. A cross-fertilization experiment was performed between *PFG*(-), *P48/45*(-) (males are infertile), and *P47*(-) (females are infertile) parasites. The rate of conversion of females to mature ookinetes is shown. F. Flow cytometry analysis of 820 parasites and *PFG*(-) parasites derived from 820 parasites. G. Volcano plot showing differential expression of genes in *PFG*(-) compared to wild-type parasites. Red dots represent female-enriched genes, and horizontal and vertical lines indicate *p*-value of 0.001 and log₂(fold change) of -2, respectively. A pie graph on the top right of the plot area shows the number of female-enriched genes among the significantly downregulated genes.

Next, we generated *PFG*(-) parasites from transgenic *P. berghei* ANKA 820 cl1m1cl1 (here called 820) parasites, which express red fluorescent protein (RFP) under the control of the female-specific promoter (*CCP2*) and GFP under the control of the male-specific promoter (dynein gene) (*PFG*(-)⁸²⁰ parasites, Fig. S1B)(12) (13). Although female gametocytes with mature morphology were observed on Giemsa-stained blood smears as the original 820 parasites, in fluorescence-activated cell sorting (FACS) analysis, no RFP-positive parasites were detected in the *PFG*(-)⁸²⁰ parasites (Fig. 2F). In contrast, GFP-positive parasites were observed in them as in the original 820 parasites (Fig. 2F). These results indicated that the promoter activity of the female-specific *CCP2* gene used for RFP expression in 820 parasites was severely reduced by the disruption of *PFG*.

Female-specific genes are globally downregulated in *PFG*(-) parasites

The above results suggest that *PFG* encodes a TF that plays an important role in female-specific gene expression. To investigate the effect of *PFG* disruption on transcriptional regulation in female gametocytes, we performed RNA-seq analyses of the wild-type and *PFG*(-) parasites (Table S1). Mice were treated with phenylhydrazine prior to the passage of parasites, and after infection, asexual stages were killed by treatment with sulfadiazine in drinking water. Three independent samples were prepared from wild-type and *PFG*(-) parasites. In *PFG*(-) parasites, 279 genes were significantly downregulated ($\log_2(\text{fold change}) < -2$ and *p*-value adjusted for multiple testing with the Benjamini-Hochberg procedure < 0.001), more than 90% of which were female-specific genes according to sex-specific transcriptome data (3) (Fig. 2G). Consistent with the results of the gene targeting experiment in 820 parasites, transcripts of the *CCP2* gene decreased over twenty times in *PFG*(-) parasites. Genes significantly downregulated in *PFG*(-) constituted more than half of the female-enriched genes (258/504, Table S1), suggesting that the expression of female-specific genes was globally downregulated in *PFG*(-) parasites.

PFG is co-localized with AP2-FG on ten-base motif

To examine whether *PFG* targeted these female-specific genes, ChIP-seq analysis was performed using *PFG::GFP* parasites and anti-GFP antibodies. In two biologically independent experiments, experiments 1 and 2, 1073 and 1,204 peaks were identified in the genome, respectively (Tables S2A and S2B), and 95% (1029 peaks) of the peaks identified in experiment 1 were common with those identified in experiment 2 (Fig. 3A). These peaks were observed upstream of the genes significantly downregulated in *PFG*(-) parasites, suggesting that *PFG* is directly involved in the transcriptional activation of these genes (examples of graphic images are shown in Fig. 3B). Statistical analysis of genomic sequences around the summits of peaks (common peaks, the same hereafter) showed that binding of *PFG* to the genome was associated with ten-base motif sequences, TGTRNNYACA (Fig. 3C), the female-specific *cis*-acting element identified in the ChIP-seq of AP2-FG (9). In comparison with the graphical views, the peaks identified in the ChIP-seq of *PFG* were co-localized with those of AP2-FG (Fig. 3B). An heat map positioned at the peak summit of *PFG* in the center showed that the position of the peaks was consistent with the ChIP-seq peaks of AP2-FG throughout the genome (Fig. 3D). These results strongly suggest that these two TFs form a complex on the ten-base motif and co-operatively activate their targets. On the other hand, graphic images also showed that some of the AP2-FG peaks lacked the corresponding ChIP-seq peaks of *PFG* (Fig. 3E).

PFG is essential for AP2-FG binding to ten-base motif

Some sequence-specific TFs form heterodimeric complexes that bind specific DNA sequences. To examine whether this was the case with *PFG* and AP2-FG, ChIP-seq analyses for each of these two TFs were performed using parasites in which the other was disrupted.

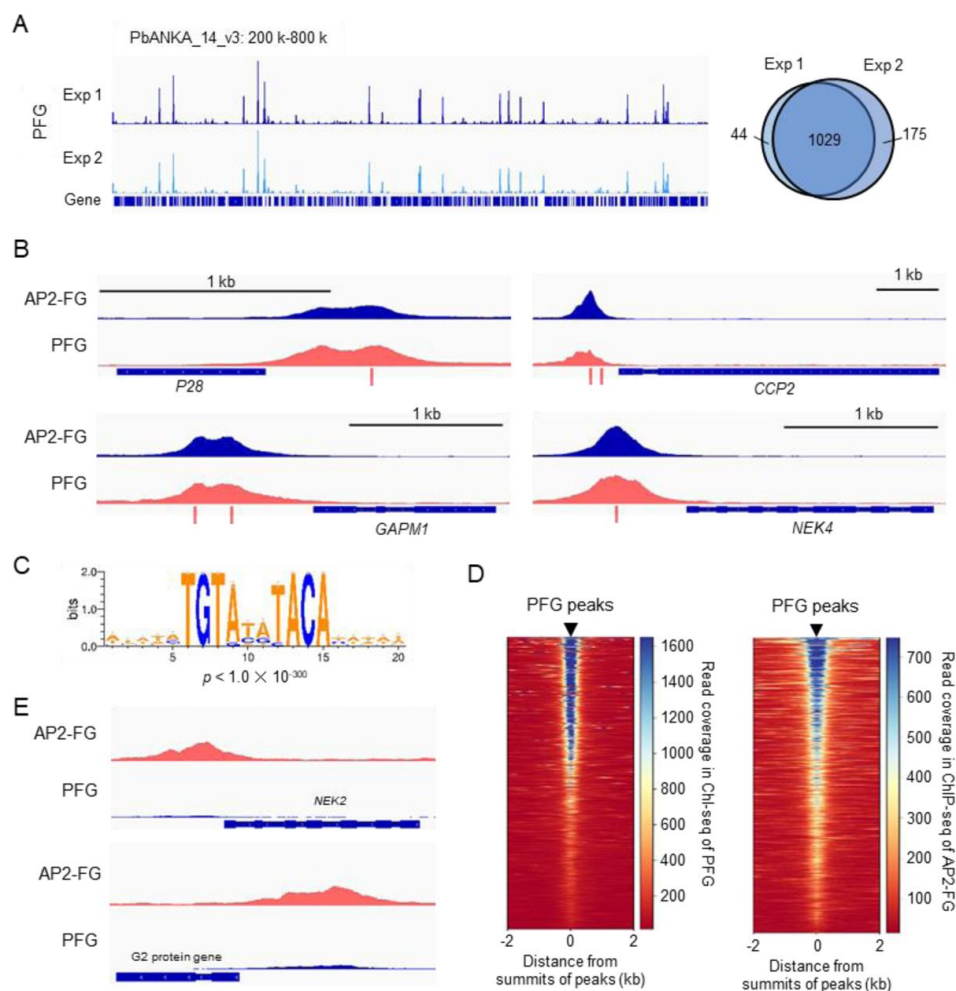


Figure 3.

PFG is co-localized with AP2-FG. A. IGV images from ChIP-seq experiments 1 and 2 of PFG on chromosome 14. Histograms show row read coverage in the ChIP data at each base. The Venn diagram on the right shows the number of peaks common in ChIP-seq experiments 1 and 2. B. Representative peak images of PFG and AP2-FG upstream of genes that were significantly downregulated in *PFG*⁻ parasites. Positions of ten-base motifs are indicated by red bars. ChIP-seq data for AP2-FG were obtained from a previous paper (23). C. The ten-base motif was enriched around the PFG peaks. Sequence logos were constructed using WebLogo 3 (<http://weblogo.threeplusone.com/create.cgi>). D. Heat maps showing coverage in ChIP-seq of PFG (left) and AP2-FG (right) with positioning summits of PFG peaks at the center. E. IGV images of ChIP-seq peaks of AP2-FG that lacked the corresponding PFG peak. Histograms show the row read coverage in the ChIP data at each base.

ChIP-seq of PFG in *PFG::GFP* parasites with the AP2-FG gene disrupted (*PFG::GFP^{AP2-FG}(-)*, Fig. S1C), 1077 and 881 peaks were obtained, and 847 peaks were common between them (Fig. 4A and Tables S2C and S2D). These peaks showed no apparent changes compared to those obtained for the original *PFG::GFP* parasites (Fig. 4B). The heat map showed that the positions of PFG peaks obtained in *PFG::GFP^{AP2-FG}(-)* parasites (common peaks, the same hereafter) were consistent with those obtained for *PFG::GFP* parasites throughout the genome, and vice versa (Fig. 4C). Statistical analysis showed that the same ten-base motif obtained in the *PFG::GFP* parasites was enriched around the summits of the PFG peak obtained in *PFG::GFP^{AP2-FG}(-)* parasites (Fig. 4D). These results indicate that PFG binds to the ten-base motif even in the absence of AP2-FG.

ChIP-seq of AP2-FG obtained in *AP2-FG::GFP* parasites with the PFG gene disrupted (*AP2-FG::GFP^{PFG}(-)*, Fig. S1D), 575 and 492 peaks were obtained, and 457 peaks were common between them (Fig. 4E and Tables S2E and S2F). The graphical appearance of the AP2-FG peaks changed drastically from that of the AP2-FG peaks obtained for the *AP2-FG::GFP* parasites. Comparison with PFG peaks showed that upon disruption of *PFG*, AP2-FG peaks common with PFG peaks disappeared, and only AP2-FG peaks not common with PFG peaks remained (Fig. 4F). In line with this, statistical analysis showed that the ten-base motif disappeared and instead two distinct motifs, GCTCA and TGCACA, became the most enriched motifs around the summits of these peaks (common peaks, the same hereafter) with a p -value of 2.1×10^{-27} and 1.3×10^{-28} , respectively (Fig. 4G). These results suggest that PFG is essential for binding AP2-FG to the ten-base motif, that is, PFG mediates AP-FG binding to the ten-base motif. The results also suggest that AP2-FG binds to the genome in two distinct forms, with and without PFG, and that the former form binds to the ten-base motif, TGTRNNYACA, and the latter binds to other motifs, supposedly GCTCA or TGCACA. In the following, we refer to the form with PFG as cAP2-FG or the complex form, and the form without PFG as sAP2-FG or the single form.

Ten-base motif is essential for binding of PFG to the genome

ChIP-seq of PFG in *PFG::GFP^{AP2-FG}(-)* parasites showed that PFG binds to the ten-base motif in the absence of AP2-FG. To demonstrate that the ten-base motif is essential for binding PFG to the genome, we performed a ChIP-qPCR assay using transgenic parasites in which the motif upstream of a target was mutated. A CPW-WPC family protein gene (PBANKA_1346300) (14) harbors a ChIP-seq peak for PFG and a ten-base motif under the summit of the peak. Three-point mutations were introduced into the motif using the CRISPR/Cas9 system, and *GFP* was fused to the PFG gene in these parasites (Fig. 4H). Introducing these mutations reduced the %input value to background levels (Fig. 4I), demonstrating that the motif is essential for PFG binding to the genome. At present, we speculate that PFG directly interacts with genomic DNA through two highly conserved regions; region 1 and region 2. However, these regions are not similar to any DNA binding domains reported thus far. In other apicomplexan orthologues, these two domains are located adjacent to one another in the protein (Fig. 1A). Therefore, these two regions may be separated by a long interval region but constitute a DNA binding domain of PFG as a result of protein folding.

AP2-FG binds to the five-base motifs directly with its AP2 domain

The results of the ChIP-seq analysis of AP2-FG in *AP2-FG::GFP^{PFG}(-)* parasites suggested that AP2-FG can bind to the genome directly through its AP2 domain. Statistical analysis showed that two distinct motifs, GCTCA and TGCACA, were enriched around the summits of ChIP-seq peaks (Fig. 4G). To investigate whether the AP2 domain of AP2-FG binds to these sequences, we performed DNA immunoprecipitation followed by high-throughput sequencing (DIP-seq) using the recombinant AP2 domain of *P. berghei* AP2-FG and *P. berghei* genomic DNA (Table S2G). Around the summits of the peaks obtained by DIP-seq analysis, the five-base motif sequence GCTCA, identical to one of motifs enriched around the summits of ChIP-seq peaks described above, was highly enriched with a p -value of 2.6×10^{-246} (Fig. 5A). In addition, two motifs, GATCA and ACTCA, which may be variants of the GCTCA motif, were also enriched, with p -value of 2.0×10^{-153}

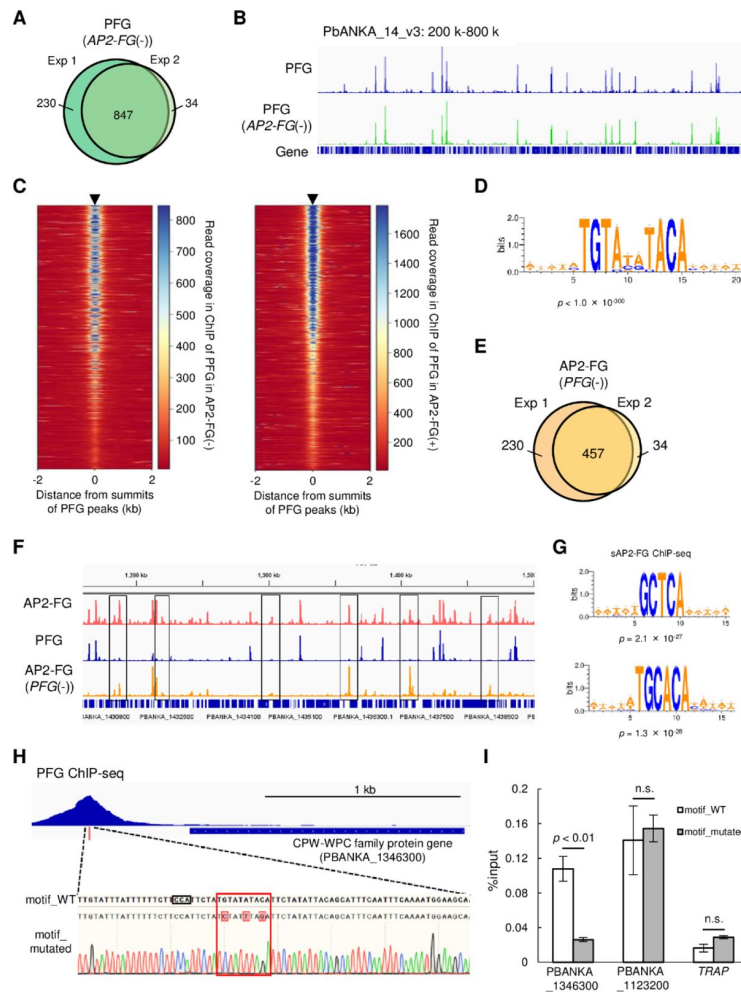


Figure 4.

PFG mediates AP2-FG binding to the ten-base motif

A. Venn diagram showing the number of common peaks between experiments 1 and 2 in ChIP-seq of PFG using *PFG::GFP^{AP2-FG(-)}* parasites. B. IGV images from the ChIP-seq peaks of PFG on part of chromosome 14. The top panel shows the ChIP peaks using *PFG::GFP* parasites, and the bottom panel shows those using *PFG::GFP^{AP2-FG(-)}* parasites. Histograms show row read coverage in the ChIP data at each base. C. Heat maps showing coverage in ChIP-seq of PFG using *PFG::GFP* parasites (left) and PFG using *PFG::GFP^{AP2-FG(-)}* parasites, with positioning summits of peaks obtained in the other ChIP-seq at the center. D. Motifs enriched around the summits of peaks identified in ChIP-seq of PFG using *PFG::GFP^{AP2-FG(-)}* parasites. The sequence logo was depicted using WebLogo 3. E. Venn diagram showing the number of common peaks between experiments 1 and 2 in ChIP-seq of AP2-FG using *AP2-FG::GFP^{PFG(-)}* parasites. F. IGV images of ChIP-seq peaks of AP2-FG and PFG in a region on which several AP2-FG peaks lack the corresponding PFG peak (chromosome 14:1,150 kb-1,500 kb). The top panel shows ChIP peaks of AP2-FG using *AP2-FG::GFP*. The middle and bottom panels show the ChIP-seq peaks of PFG using *PFG::GFP* and the ChIP peaks of AP2-FG using *AP2-FG::GFP^{PFG(-)}* parasites, respectively. The AP2-FG peaks lacking their corresponding PFG peaks are highlighted by rectangles. Histograms show row read coverage in the ChIP data at each base. G. Motifs enriched around the summits of peaks identified in the ChIP-seq of AP2-FG using *AP2-FG::GFP^{PFG(-)}* parasites. Sequence logos were depicted using WebLogo 3. H. Peak image of PFG and the ten-base motif under the summit upstream of a CPW-WPC family protein gene (PBANKA_1346300). The motif was mutated using the CRISPR/Cas9 system, and the mutation was confirmed by Sanger sequencing (the lowest panel; mutated nucleic acid residues are highlighted). The protospacer adjacent motif (PAM) sequence used for targeting is also highlighted by a rectangle. I. ChIP-qPCR analysis of PFG in the upstream region of PBANKA_1346300. Gray and white bars indicate the results using wild-type and motif-mutated parasites, respectively. Three independent biological experiments were performed. The results are shown as a percentage input. Error bars indicate standard error. Experiments with another CPW-WPC family protein gene (PBANKA_1123200) and TRAP were performed as positive and negative controls, respectively.

and 6.8×10^{-51} , respectively (Fig. 5A). In contrast, the six-base motif, TGCACA, was not enriched around DIP-seq peaks (p -value = 0.051), suggesting that it is not a binding motif of AP2-FG. Two minor motifs, GATCA and ACTCA, were also enriched around the summits of ChIP-seq peaks of sAP2-FG, with p -value of 2.0×10^{-53} and 1.0×10^{-13} , respectively. In the ChIP-seq of sAP2-FG, 84.2% of the peaks contained at least one of these five-base motifs within 300 bp from the summits, and the average distance from the summits was 62.4 bp (Fig. 5B). The most enriched motif matched well with the binding sequence of the AP2 domain of *P. falciparum* AP2-FG, which was reported by Campbell *et al.* (15). Collectively, these results suggested that AP2-FG binds directly to these motifs through its AP2 domain.

A ChIP-qPCR assay was performed to confirm that this motif is essential for the binding of sAP2-FG to the upstream of target genes. NEK2 (PBANKA_1240700) is a Nima-related protein kinase essential for parasite transmission to mosquito vectors (16). The NEK2 gene is one of the target genes of sAP2-FG, harboring one five-base motif sequence under the ChIP-seq peak. We prepared parasites with this motif mutated from AP2-FG::GFP^{PFG}(-) parasites (Figs 5C) and performed a qPCR assay with anti-GFP antibodies. Upon addition of the mutation, the AP2-FG associated with the locus decreased to background levels (Fig. 5D). These results demonstrated that AP2-FG binds to the five-base motif through its AP2 domain.

Five-base motif acts as a cis-activating element on the promoter

The function of the five-base motif as *cis*-acting element in the female-specific promoter was determined by promoter assays using the centromere plasmid (17). The upstream region of the female-specific gene *P28* was used in this assay (Fig. 5E). The region harbors one ten-base motif and three five-base motifs. The mCherry gene on the centromere plasmid was expressed under the control of this promoter region, and mCherry expression was analyzed by FACS (Fig. 5F). Promoter activity was reduced by introducing mutations to the ten-base motifs, but the signals were still retained. The signals became almost undetectable by introducing mutations into the five-base motifs. These results demonstrate that the five-base motifs are female-specific *cis*-acting elements and function independently of the ten-base motif in the promoter.

Genome-wide identification of cAP2-FG and sAP2-FG targets

Our ChIP-seq analyses suggested that the targets of AP2-FG were constituted by targets of cAP2-FG and sAP2-FG. To investigate which genes were commonly or uniquely regulated by these two forms of AP2-FG, we predicted the targets of cAP2-FG and sAP2-FG separately. As targets of cAP2-FG, 810 genes were predicted from ChIP-seq peaks of PFG obtained from PFG::GFP parasites (common peaks of two experiments, the same hereafter) (Fig. 6A and Table 3A). These targets broadly contained genes related to female gametocyte-specific functions, including genes for female gametocyte development, such as genes for egress from the erythrocyte and fertilization, and genes for zygote/ookinete development, such as meiosis-related genes, genes for pellicular and subpellicular structures, ookinete surface or secretory protein genes, and crystalloid protein genes (Table S3A). As target genes for sAP2-FG, 321 were predicted by ChIP-seq in AP2-FG::GFP^{PFG}(-) parasites (Fig. 6B and Table S3B). Targets of sAP2-FG also contained many genes for female gametocyte-specific functions, but in smaller numbers than those in cAP2-FG targets. As a result of these target predictions, 967 genes were identified as targets for either cAP2-FG or sAP2-FG, and 164 genes were common between them (Fig. 6C). Based on this classification, the cAP2-FG and sAP2-FG targets had the following features: First, genes known to be involved in gametocyte development are target genes of sAP2-FG, and most of them are also targets of cAP2-FG (thus belonging to common targets). Second, genes for zygote/ookinete development are targets of cAP2-FG, and some are also targets of sAP2-FG. Among these, most meiosis-related genes and crystalloid protein genes are the target genes unique to cAP2-FG.

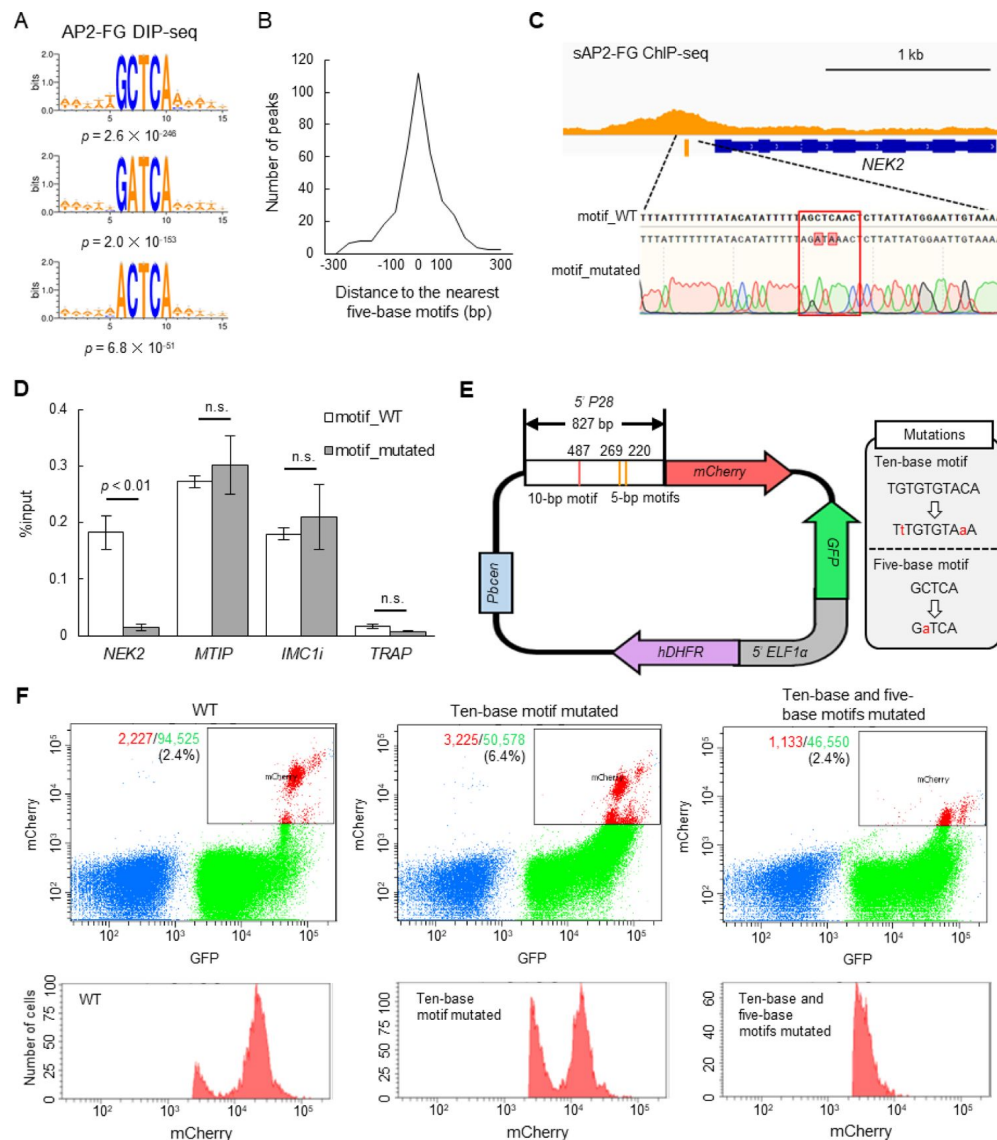


Figure 5.

sAP2-FG binds to five-base motifs that function as a *cis*-acting element independent of the ten-base motif.

A. Motifs enriched around peaks identified in DIP-seq of the AP2 domain of AP2-FG. Sequence logos were depicted using WebLogo 3. B. Histogram showing the distance between the peak summit and the nearest five-base motifs in (A) for each ChIP peak of sAP2-FG. C. Peak image of sAP2-FG upstream of *NEK2* and the five-base motif under the summit. The motif was mutated, and the sequence was confirmed by Sanger sequencing (lowest panel). The protospacer adjacent motif (PAM) sequence used for targeting is highlighted by a rectangle. Mutated nucleic acid residues have also been highlighted. D. ChIP-qPCR analysis of sAP2-FG in the upstream region of *NEK2*. Gray and white bars indicate the results obtained for wild-type and motif-mutated parasites, respectively. The results are shown as mean %input values from three biologically independent experiments. Error bars indicate standard error. Experiments with *MTIP* and *IMC1i* were performed as positive controls, and that in *TRAP* as a negative control. E. Schematic diagram of a *Plasmodium* centromere plasmid used to assess promoter activity of *P28*. Mutations introduced into the ten-base and five-base motifs of the *P28* promoter are described in a grey box. *Pbcen*, a sequence of the *P. berghei* chromosome 5 centromere, 5'-ELF1a, a bidirectional promoter of the elongation 1a gene for conferring constitutive expression F. FACS analysis of parasites harboring the *P28*-reporter plasmid (E). Parasites were gated on forward-scatter and staining with Hoechst 33342, and then on GFP. The percentages of mCherry-positive parasites in all gated cells are shown in the left panel. The histogram on the right shows the number of mCherry-positive parasites at different signal strengths.

Target prediction also showed that *PFG* is a target of sAP2-FG and cAP2-FG. Upstream of *PFG* harbors the binding sites of three different TFs: AP2-G, sAP2-FG (AP2-FG), and cAP2-FG (PFG) (Fig. 6D). These results suggest that *PFG* is activated in early females by sAP2-FG and then by cAP2-FG in later female gametocytes. Meanwhile, the zygote TF AP2-Z was predicted to be a target of cAP2-FG only (18). Together with the results obtained in previous studies (8)(18), a cascade of TFs starting from AP2-G could be summarized as in Fig. 6E.

Impact of *PFG* disruption on the expression of AP2-FG target genes

To examine the independence of transcriptional activation by cAP2-FG from that of sAP2-FG, the impact of *PFG* disruption on the expression of AP2-FG target genes was assessed separately in the three groups described in Fig 6C, that is, unique for sAP2-FG, unique for *PFG*, and common for both, using RNA-seq data already presented in Fig. 2G. Among the 279 genes significantly downregulated in *PFG*(-) parasites, 165 genes were targets for *PFG* (unique for *PFG* or common for sAP2-FG and *PFG*), and only four genes were targets unique to sAP2-FG. The log2 distribution (fold change = *PFG*(-)/wild type) in the three groups of target genes showed that the average value was significantly higher (i.e., less down-regulated) in targets unique to sAP2-FG than in the other two groups (targets unique to cAP2-FG or common targets for both), with p-values of 1.3×10^{-10} and 1.4×10^{-5} , respectively, by two-tailed Student's t-test (Fig. 6F). In addition, the average log2 (fold change) value of the common target genes was relatively higher (i.e., less down-regulated) than that of targets unique to *PFG*, suggesting that transcriptional activation by sAP2-FG partly complements the impact of *PFG* disruption on these common target. These results suggest that the trans-activity of cAP2-FG and sAP2-FG is independent of each other.

Discussion

Genes transcribed in female gametocytes are involved in various steps of parasite transmission to the mosquito vector, including sex-specific differentiation in the vertebrate host and zygote/ookinete development that take place in the mosquito vector. In a previous study, we reported that this wide range of gene expression repertoires in females is regulated directly by a female-specific TF AP2-FG that binds to a ten-base *cis*-acting element upstream of these genes (9). In this study, we report two findings that change our previous view. First, the binding of AP2-FG to the ten-base motif is mediated by another female-specific TF, *PFG*. Second, there is another *cis*-acting element in female-specific promoters, and AP2-FG binds to this element directly as a *trans*-activating factor. These findings suggest that transcriptional regulation of female-specific genes is not as simple as previously thought but is regulated by various promoters composed of different combinations of the two *cis*-acting elements. Considering the different expression profiles of *PFG* and AP2-FG, it is assumed that different expression profiles are generated among female-specific genes by the combination of these two *cis*-acting elements.

The present study suggests that *PFG* is upregulated in two steps in female gametocytes; first by sAP2-FG, and then by cAP2-FG. Transcriptional activation of *PFG* by cAP2-FG constitutes a transcriptional positive feedback loop, suggesting that after activation by sAP2-FG, *PFG* maintains its transcription through this autoactivation mechanism. As the expression of *PFG* increases via this mechanism, AP2-FG recruited by *PFG* (cAP2-FG) increases and eventually becomes predominant in the transcriptional regulation of female gametocytes. Fig. 6G illustrates this model. Considering that targetome of cAP2-FG contains many ookinete genes, transcripts for ookinete development would gradually increase in transcriptome of females. In this model it is hypothesized that binding of AP2-FG contributes to the trans-activity of cAP2-FG. This is supported by the observation that expression of many unique targets of *PFG* decreases in *AP2-FG*-disrupted parasites (9). However, it remains elusive how AP2-FG contributes to this activity. As discussed below, *PFG* may have its own trans-activity for the ten-base element. It is necessary to investigate the role of AP2-FG in cAP2-FG as a next step.

In *AP2-FG*-disrupted parasites, while female gametocytes display immature morphologies, a few develop into retort-form ookinetes. Because PFG is essential for the parasite's capacity to produce ookinetes, this phenotype suggests that PFG retains trans-activity in the absence of *AP2-FG* and partly promote ookinete development. The observation also supports this assumption that RFP signals in female gametocytes could still be detected in *AP2-FG(-)*⁸²⁰ parasites by FACS (9) even though RFP signals were reduced to undetectable level in *PFG(-)*⁸²⁰ parasites under the same experimental conditions. This suggests that the promoter of the *CCP2* gene, which is a target of PFG only, is still active in *AP2-FG(-)*⁸²⁰ parasites. This putative trans-activity of PFG could also explain why the expression of *PFG* was still observed in *AP2-FG*-disrupted parasites. It is assumed that PFG continues to activate its transcription through a positive feedback loop once it is induced by *AP2-G* and complements the absence of *AP2-FG*. Based on this assumption and the model presented in Fig. 6G, phenotypic differences between *PFG*-disrupted and *AP2-FG*-disrupted parasites can be explained as follows. *PFG(-)* parasites completely lose their ability to develop into ookinetes, because the expression of genes necessary for ookinete development is significantly reduced. However, they manifest mature morphologies in females owing to the induction of the genes necessary for female development by *sAP2-FG*. In contrast, *AP2-FG(-)* parasites show immature morphologies in females because the expression of genes involved in female development is downregulated in the early stage. Despite this immature morphology, they retain the ability to fertilize and develop into retort-form ookinetes because PFG is still expressed in *AP2-FG(-)* and activates genes for zygote/ookinete development. *P. falciparum* gametocytes require 9–12 days to mature, which is much longer than that of *P. berghei* (19). Meanwhile, it has been reported that the ten-base motif is highly enriched in the upstream regions of gametocyte-specific genes also in *P. falciparum* (20). Thus, despite the difference in maturation periods, PFG is likely to play an important role in the transcriptional regulation of female gametocyte development in *P. falciparum*.

In conclusion, our results suggest that the two forms of *AP2-FG*, which correspond to distinct female-specific *cis*-acting elements, play pivotal roles in female gametocyte development and can generate variations among female-specific promoters. The present findings provide the possibility to predict the expression profile of each gene by analyzing its promoter sequence and further estimating its functions according to this regulation. In the next step, it will be necessary to compare the properties of the number of actual promoter activities *in vivo* and improve the model presented in this study. Such studies deepen our understanding of the sexual stage, including gametocyte development in erythrocytes, fertilization, meiosis, and development into ookinetes, and provide clues to interrupt parasite transmission from humans to mosquitoes.

Materials and Methods

Parasite preparations

The ANKA strain of *P. berghei* was maintained in female BALB/c mice (6–10 weeks of age). To examine the number of oocysts, infected mice were exposed to *Anopheles stephensi* mosquitoes. Fully engorged mosquitoes were selected and maintained at 20 °C. The number of oocysts was evaluated 14 days after the infective blood meal. To prepare mature schizonts, infected mouse blood was cultured in the medium for 16 h. Mature schizonts were purified using Nicoprep density gradient. To prepare gametocyte-enriched blood, mice were pretreated with hydrazine and infected with *P. berghei* by intra-abdominal injection of infected blood. After parasitemia increased to over 1%, they were treated with sulfadiazine for two days in drinking water (20 mg/L) to kill asexual parasites. After checking the exflagellation rate, whole blood was collected.

Preparation for transgenic parasites

Parasite transfection was performed as previously described (9). Briefly, cultured mature schizonts were purified using a density gradient of Nicoprep, transfected with DNA constructs by electroporation, and injected intravenously into mice. These mice were treated with pyrimethamine to select the parasite integrated with the construct by double crossover homologous recombination. The parasites were cloned by limiting dilution to obtain transgenic parasite clones. To prepare the targeting construct, DNA fragments for recombination were annealed to each side of the DNA fragments containing *human DHFR* as a selectable marker gene by overlapping PCR. Constructs for parasites expressing the GFP-fused genes were prepared as follows. Briefly, DNA fragments corresponding to the 3' portion of the gene were subcloned into the plasmid vector containing the GFP gene, the 3'-terminal portion of the *P. berghei* HSP70 gene, and a selectable marker cassette for expressing the *human DHFR* gene. The plasmid was separated from the vector backbone by digestion with restriction enzymes NotI and BanHI.

Transgenic parasites with mutated motif sequences were prepared using the CRISPR/Cas9 method, as reported previously (21). Briefly, cultured mature merozoites of CAS9-expressing *P. berghei* parasites were transfected with a gRNA plasmid vector and a linear DNA template containing mutations in the motif sequences and injected intravenously into mice. These mice were treated with pyrimethamine for two days to deplete parasites in which the locus did not change. The parasites were cloned by limiting dilution. Mutations in these motifs were confirmed by direct genome sequencing.

Flow cytometric analysis

Flow cytometric analysis was performed using an LSR-II flow cytometer (BD Biosciences). In experiments using 820 parasites, the tail blood from infected mice was selected via gating with forward scatter and staining with Hoechst 33342 (excitation = 355 nm, emission = 450/50). The gated population was then analyzed for GFP fluorescence (excitation = 488 nm, emission = 530/30) and RFP fluorescence (excitation = 561 nm, emission = 610/20). In the promoter assay (using parasites transfected with a centromere plasmid), the tail blood from infected mice was selected via gating with forward scatter and staining with Hoechst 33342 (excitation = 355 nm, emission = 450/50), followed by GFP fluorescence (excitation = 488 nm, emission = 530/30). The gated population was analyzed for mCherry fluorescence (excitation = 561 nm, emission = 610/20). Analysis was performed using the DIVER program (BD Biosciences).

ChIP-Seq

Two infected mice were used in each ChIP-seq experiment. ChIP-seq experiments were performed as previously described (22). The gametocyte-enriched blood was filtered using a Plasmodipur filter to remove white blood cells and fixed in 1% paraformaldehyde for 30 min with swirling. Red blood cells were lysed in 0.84% NH_4Cl , and residual cells were lysed in a lysis buffer containing 1% SDS. The lysate was sonicated in Bioruptor 2. After centrifugation, the supernatant was diluted with dilution buffer, and chromatin was immunoprecipitated with Dynabeads protein A coated with an anti-GFP antibody. DNA fragments were recovered from the precipitated chromatin and used for library preparation for sequencing. The library was prepared using a Hyper Prep Kit (KAPA Biosystems). Sequencing was performed on an Illumina NextSeq sequencer.

Analysis of ChIP-seq data

Sequence data were mapped onto the *P. berghei* genome sequence (PlasmoDB, version 3) using BOWTIE2 software with default settings. Mapping data were analyzed using the MACS2 peak-calling algorithm. Conditions for peak calling were $\text{FDR} < 0.01$ and fold enrichment > 2 or 3. To detect large peaks constituted from multiple peaks, the option "Callsummit" was used. Mapped read data were visualized using IGV software. Sequences concentrated around the predicted summits of ChIP-seq peaks were investigated with Fisher's exact test carried out between 200-bp

regions that had summits in the center and 200-bp regions excised from the genome, excluding the former regions, to cover the entire genome sequence. Sequences with the least p-values were combined with the common sequence motifs. Genes were determined as targets when their 1.2-kbp upstream regions harbored the predicted summits of ChIP-seq peaks. When the upstream intergenic region was less than 1.2 kbp, the entire intergenic region was used for target prediction. Two biologically independent ChIP-seq experiments were performed. The ChIP-seq data were deposited in the Gene Expression Omnibus (GEO) with the accession number GSE226028.

DIP-seq

The recombinant AP2 domain of AP2-FG was prepared as a GST-fused protein using the same procedure as reported previously⁽⁸⁾. DIP was performed as reported previously⁽⁸⁾. Briefly, recombinant GST-fused AP2 domain of AP2-FG and genomic DNA fragments were incubated at room temperature for 30 min in 100 μ L of binding/washing buffer. The mixture was then incubated with 50 μ L glutathione sepharose resin (Cytiva) at room temperature with rotation. Resin was washed three times with 150 μ L of binding/washing buffer, and bound protein was eluted with 10 mM glutathione solution. Recovered DNA (5 ng) was sequenced using the same procedures as in ChIP-seq. Input genomic DNA fragments were also sequenced as a control. Analysis of the sequence data was performed using the same procedures as in ChIP-seq. Data were deposited in the GEO database under accession number GSE226028.

ChIP-qPCR

ChIP for ChIP-qPCR was performed using gametocyte-enriched mouse blood, using the same procedures as in ChIP-seq. One infected mouse was used in each ChIP experiment. qPCR was performed using the TB Green Fast qPCR Mix (Takara Bio) according to the manufacturer's protocol. Three independent ChIP experiments were performed, and the percentage of input DNA (DNA extracted from lysate before ChIP) was compared between *P. berghei* parasites expressing GFP-fused TF and those with the mutated motifs. The primers used for qPCR are listed in Table S4.

RNA-seq analysis

Gametocyte-enriched blood was passed through a filter and subjected to erythrocyte lysis in 0.84% NH_4Cl . According to the manufacturer's protocols, total RNA was extracted from residual cells using Isogen II (Nippon Gene). Libraries were prepared using the RNA Hyperprep kit (KAPA Biosystems) and sequencing was performed using an Illumina NextSeq sequencer.

Analysis of RNA-Seq data

Three independent experiments were performed using *PFG*-disrupted parasites. Read data were counted using FeatureCount software. RPKM (Reads Per Kilobase of exon per million mapped reads) was calculated for each gene, and only genes with an RPKM maximum > 20 in wild-type parasites were used for the following analyses. Additionally, genes located in the sub telomeric regions were excluded because of their variable expression among the clones. The ratio of read numbers and padj was calculated for each gene between wild-type and *gSNF2*-disrupted parasites using the DEseq2 software. Volcano plots of the obtained results were created using DDplot2 software. The RNA-seq data were deposited in GEO with the accession number GSE226028. The RNA-seq data deposited in GEO with accession number GSE198588 were used as data for the wild-type parasites.

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Supporting information

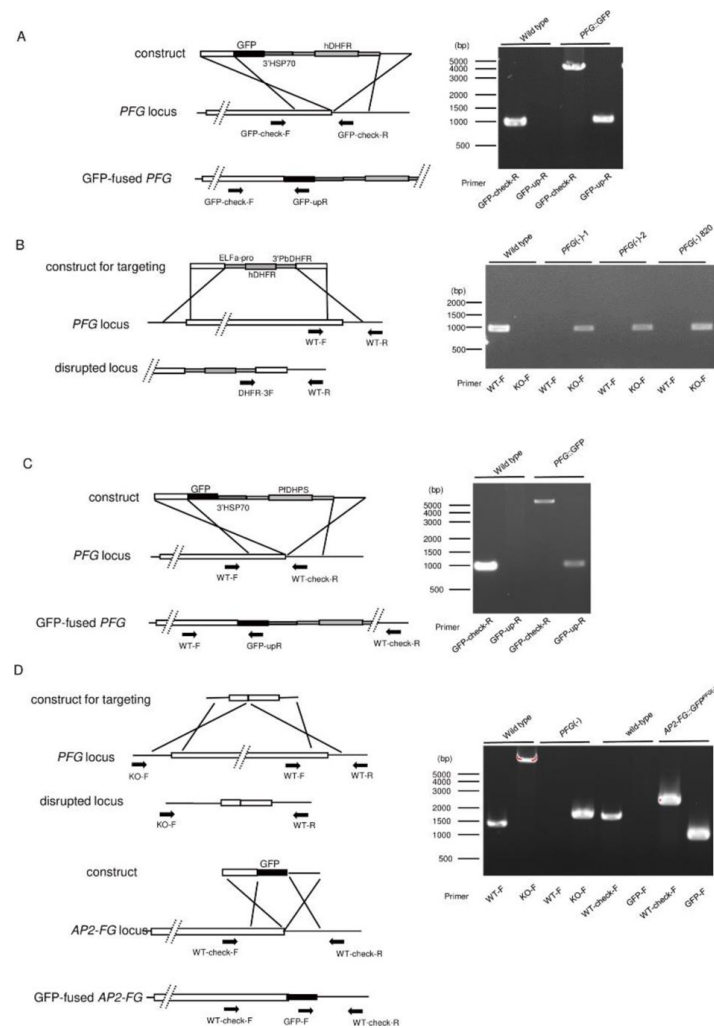


Fig. S1.

Preparation of transgenic parasites.

A. Preparation of *PFG::GFP* parasites. B. Preparation of *PFG(-)* and *PFG(-)⁸²⁰* parasites. C. Preparation of *PFG::GFP^{AP2-FG(-)}* parasites. *PFG::GFP^{AP2-FG(-)}* parasites were prepared from *AP2-FG(-)* parasites using the sulfadiazine-resistant dihydropteroate synthase (DHPS) gene as a selectable marker gene. D. Preparation of *AP2-FG::GFP^{PFG(-)}* parasites. *AP2-FG::GFP^{PFG(-)}* parasites were prepared using CRISPR/Cas9 in two steps. First, *PFG* was disrupted in *pbcas9*, and then *GFP* was fused to the 3'-portion of the *AP2-FG* gene.

Captions of supplementary tables

Table S1 (separate file). Differential expression analysis between wild-type and *PFG(-)* parasites.

Table S2 (separate file). Peaks identified in ChIP-seq experiments and their nearest genes.

Table S2A. Peaks identified in ChIP-seq of PFG with their nearest genes (experiment 1).

Table S2B. Peaks identified in ChIP-seq of PFG (experiment 2).

Table S2C. Peaks identified in ChIP-seq of PFG using *AP2-FG(-)* parasites (experiment 1)

Table S2D. Peaks identified in ChIP-seq of PFG using *AP2-FG(-)* parasites (experiment 2).

Table S2E. Peaks identified in ChIP-seq of *sAP2-FG* (experiment 1).

Table S2F. Peaks identified in ChIP-seq of *sAP2-FG* (experiment 2).

Table S2G. Peaks identified in DIP-seq using recombinant AP2 domain of *AP2-FG* and *P. berghei* genomic DNA.

Table S3 (separate file). Target genes predicted from ChIP-seq peaks. Table S3A. PFG target genes predicted from ChIP-seq peaks.

Table S3B. *sAP2-FG* target genes predicted from ChIP-seq peaks.

Table S4. List of primers used in this study.

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Reviewer #1 (Public Review):

Gametocytes are erythrocytic sexual stages of the malaria-causing parasite *Plasmodium*, and are essential for parasite transmission and reproduction in the mosquito vector. In this study, Murata et al investigated the mechanisms of gene regulation in female gametocytes in the rodent malaria model parasite *Plasmodium berghei*. According to current views, gene regulation in *Plasmodium* parasites is dominated by the family of AP2 transcription factors (TFs), such as the AP2-G TF, which drives sexual commitment. The same authors previously identified one AP2 TF, called AP2-FG, as an essential TF mediating differentiation of female gametocytes. Here, they identified a novel protein, called PFG (for partner of AP2-FG, also described as Fd2 in a recently published study), which cooperates with AP2-FG to regulate a subset of female gametocyte genes.

PFG was identified among AP2-G targets, but possesses no known DNA binding or other characterized domain. The authors show that PFG-knockout *P. berghei* parasites can form

male and female gametocytes yet cannot transmit to mosquitoes, due to a defect in female gametocyte development. Using RNA-seq, they show that many female-specific genes are down-regulated in PFG(-)parasites. Chromatin immunoprecipitation combined with DNA sequencing (ChIP-seq) revealed that PFG colocalizes with AP2-FG on a ten-base motif that is enriched upstream of female-specific genes. Importantly, the ChIP-seq profile of PFG is unchanged in the absence of AP2-FG, suggesting that PFG binds to DNA independently of AP2-FG. Mutation of the ten-base motif in one target gene using CRISPR-Cas9 demonstrates that this motif is required for PFG localization at the gene locus. The data also show that binding of AP2-FG is affected in the absence of PFG, with disruption of AP2-FG interaction with the ten-base motif, but conservation of AP2-FG binding to distinct five-base motifs. Using a recombinant AP2 domain from AP2-FG, the authors demonstrate that the AP2 domain of AP2-FG binds to the five-base motifs. Using CRISPR they show that disruption of the five-base motifs in the genome abrogates AP2-FG binding, and using a reporter expression system they confirm that these motifs act as a cis-activating promoter element.

Through the analysis of target genes based on the presence of the ten- versus five-base motifs, the authors propose a model where AP2-FG can function in two forms, associated or not with PFG, and acting on the ten- or five-base motifs, respectively, to regulate distinct gene subsets during development of female gametocyte development.

The paper is well written, with a clear narrative, and the work is very well performed, relying on robust molecular approaches. Generally the conclusions and the model proposed by the authors are well supported by the data. Nevertheless, the study as it stands raises a number of questions. While the data convincingly show that PFG and AP2-FG cooperate to regulate the expression of a subset of female-specific genes, the paper does not show whether the two proteins actually interact with each other to form a complex. Also, how PFG binds to DNA and whether the protein has transactivating activity remains elusive, as the protein apparently possesses no known DNA-binding or activating domain. These points could be discussed in more detail in the manuscript and/or be the subject of follow up studies.

In summary, this work reveals the essential role of a Plasmodium protein with no known DNA binding or regulatory domain, illustrating that unknown facets remain to be uncovered in this fascinating pathogen.

- <https://doi.org/10.7554/eLife.88317.2.sa2>

Reviewer #2 (Public Review):

Murata et al have characterized a transcription activator previously identified in an earlier genetic screen by Russell et al (named Fd2; for female-defective 2), here named PFG. The authors show solid evidence that PFG is a partner of the previously described transcription factor AP2-FG and describe three sets of genes: genes activated by PFG or AP2-FG alone and genes activated by the complex. The authors also show differential binding to the target DNA sequences by AP2-FG to either a 10bp, if in a complex with PFG or a 5bp motif if alone. In all, this is a useful study which further elucidates the underlying regulatory network that drives development of sexual stages and ultimately transmission to mosquitoes. The data presented are clear and solid and the conclusions drawn are mostly supported by the results shown.

A few comments:

Given that the transcriptional programme is so dynamic, the timing of the ChIP-seq experiments is crucial. Could the authors clarify the timings of the different ChIP-seq experiments (AP2-FG, PFG, PFG in AP2-FG-, AP2-FG in PFG-, ...)

Fig 4c is an example of great overlap of peaks, but it would be helpful if the authors could quantify the overlaps between experiments (and describe the overlap parameters used).

It remains unclear if AP2-FG and PFG interact directly or if they bind sequentially in the transcriptional activation process. Perhaps they are part of a larger complex? Immunoprecipitation followed by mass spectrometry of the GFP-tagged version of PFG in the presence and absence of AP2-FG would be highly informative.

- <https://doi.org/10.7554/eLife.88317.2.sa1>

Reviewer #3 (Public Review):

This study is well designed and executed and provides new and important insights into the role of two TFs during the maturation of female gametocytes and fertilization in the mosquito midgut. However, it but would benefit from a more thorough characterization of the phenotype to understand at which step of development these factors are required.

Overall the authors have shown only limited willingness to comprehensively address reviewer concerns and incorporate their suggestions.

- <https://doi.org/10.7554/eLife.88317.2.sa0>

Author Response

The following is the authors' response to the original reviews.

Reviewer #1 (Recommendations For The Authors):

The manuscript is very well written, the data are clearly presented and the methodology is robust. I only have suggestions to improve the manuscript, to make the study more appealing or to discuss in more detail some questions raised by the work.

1. *In the study as it stands, PFG seems to come out of the blue. The authors apparently selected this protein based on sequence conservation between species but this is unlikely to be sufficient to identify novel TFs. Explaining in more detail the reasoning that led to PFG would make the story more appealing. Perhaps PFG was identified through a large reverse genetics screening?*

Response: Thank you for your suggestion. We identified this gene solely by the strategy we described in the manuscript. We decided on this strategy based on the findings of our previous study on AP2-Family TFs, whose DNA binding domains are highly conserved among Plasmodium orthologues. Using this screening strategy, we identified a novel AP2 family TF AP2-Z. The results of the present study demonstrated that this strategy is applicable to TFs other than those belonging to the AP2 family. We are aware that this strategy is not all-encompassing. In fact, we failed to identify HDP1 as a candidate TF when it was also in the target list of AP2-G. However, at present, this is our primary strategy for identifying novel TFs in the targetome.

1. *The authors propose that PFG and AP2-FG form a complex, but this is actually not shown. Did they try to document a physical interaction between the two proteins, for example using co-IP?*

Response: Even when the two molecules were identified to be at the same position by ChIPseq, it cannot be concluded that they form a physical complex because it is possible that they competitively occupy the region. However, in this study, we performed ChIP-seq in the absence of PFG and demonstrated that the cAP2-FG peaks disappeared while those of sAP2-FG

remained. This result can only be explained by the two proteins forming a complex at this region, which excludes the possibility that AP2-FG binds the region independently.

1. It is unclear how PFG can bind to DNA in the absence of DNA-binding domain. Did the authors search for unconventional domains in the protein? This should be at least discussed in the manuscript.

Response: We speculate that the two highly conserved regions, region 1 and region 2, function as DNA-binding domains in PFG. However, this domain is not similar to any DNA binding domains reported thus far. A straightforward way to demonstrate this would be to perform in vitro binding assays using a recombinant protein. However, thus far, we have not succeeded in obtaining soluble recombinant proteins for these regions. We have added the following sentences to the results section.

“At present, we speculate that PFG directly interacts with genomic DNA through two highly conserved regions; region 1 and region 2. However, these regions are not similar to any DNA binding domains reported thus far. In other apicomplexan orthologues, these two domains are located adjacent to one another in the protein (Fig. 1A). Therefore, these two regions may be separated by a long interval region but constitute a DNA binding domain of PFG as a result of protein folding.”

1. How do the authors explain that PFG is still expressed in the absence of AP2-FG? Is AP2G alone sufficient to express sufficient levels of the protein? Is PFG down-regulated in the absence of AP2-FG?

Response: Our previous ChIP-seq data indicate that PFG is a target of AP2-G. According to the study by Kent et al. (2018), this gene is up-regulated in the early period following conditional AP2-G induction. The results of the present study showed that PFG is capable of autoactivation through a transcriptional positive feed-back loop. These results suggest that PFG can maintain its expression to a certain level once activated by AP2-G, even in the absence of AP2-FG. In our previous microarray analysis, significant decreases in PFG expression were not observed in AP2-FG-disrupted parasites.

1. How do AP2-FG regulated genes (based on RNAseq) compare with the predicted cAP2FG/sAP2-FG predicted genes (based on ChIPseq)? Are the two subsets included in the genes that are actually down-regulated in AP2-FG(-)?

Response: Disruption of the AP2-FG gene impairs gametocyte development. We considered that the direct effect of this disruption would be difficult to analyze in gametocyte-enriched blood, in which gametocytes are pooled during sulfadiazine treatment to deplete asexual stages. Therefore, in our previous paper, we performed microarray analysis between WT and KO parasites to detect the direct effect of AP2-FG disruption on target gene expression, using mice which were synchronously infected with parasites. According to our results, 206 genes were down-regulated in AP2-FG-disrupted parasites. Of these genes, 40 and 117 were targets of sAP2-FG and cAP2-FG, respectively. However, it is still possible that a significant proportion of genes were indirectly down-regulated by AP2-FG disruption, which may impair gametocyte development. Moreover, based on the results of the present study, expression of a significant proportion of AP2-FG target genes could be complemented by PFG transcription. We believe that it would be difficult to compare the direct effects of these TFs on gene expression via transcriptome analysis (therefore, targetome analysis is important). In this study, we compared the expression of target genes of sAP2-FG and cAP2FG between PFG(-) and WT parasites. We expected that down-regulation of PFG (cAP2FG) targets would be complemented with transcription by sAP2-FG.

1. Minor points

-Page 5 Line 10, remove "as"

Response: We have corrected this.

-Page 7 Lines 4-13: *is it possible to perform the assay in PFG(-) parasites?*

Response: Thank you for your question. Even when the marker gene expression was decreased in PFG(-) parasites, we cannot conclude the reason to be a direct effect of the mutation. To determine the function of the motif, it is necessary to perform the assay using wild-type parasites.

-Page 7 Line 45: *Fig6C instead of 5C*

Response: Thank you for pointing this out. We have corrected this.

-Page 8 Line 27: *"decreases"*

Response: Thank you for pointing this out. We have corrected this.

-Page 8 Line 36: *PFG instead of PGP*

Response: We have corrected this.

-Page 8 Line 39: *remove "the fact"*

Response: We have removed this word.

-Page 8 Line 42: *Fig6G instead of 5G*

Response: We have corrected this.

-Page 8 Line 43: *PFG instead of PGP*

Response: We have corrected this.

-Page 9 Line 23: *"electroporation"*

Response: We have corrected this.

-Page 9 Line 32: *"BamHI"*

Response: We have corrected this.

-Fig 2E: *in the crosses did the authors check oocyst formation in the mosquito?*

Response: We did not check oocyst formation because abnormalities in males may not affect oocyst formation.

-Page 17, legend Fig3, Line 14, *there is probably an inversion between left and right for PFG versus AP2-FG (either in the legend or in the figure)*

Response: Thank you for pointing this out. PFG peaks are located in the center in both heat maps. The description “AP2-FG peaks” over the arrowhead in the left map was incorrect. We have corrected this to “PFG peaks”. The peaks in the left heat map must be located in the center; thus, this figure might be redundant.

Reviewer #2 (Recommendations for the Authors):

- *Could the authors please state in the results section that PFG stands for partner of AP2FG.*

Response: Thank you for the comment. We have added the following to the results section:

“Through this screening, a gene encoding a 2709 amino acid protein with two regions highly conserved among Plasmodium was identified (PBANKA0902300, designated as a partner of AP2-FG (PFG; Fig. 1A).”

- *Given that the transcriptional program is so dynamic, the timing of the ChIP-seq experiments is crucial. Could the authors clarify the timings of the different ChIP-seq experiments (AP2-FG, PFG, PFG in AP2-FG-, AP2-FG in PFG-, ...)*

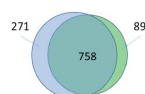
Response: Thank you for the comment. To deplete any parasites in the asexual stages, all ChIP-seq experiments in this study were performed using blood from mice treated with sulfadiazine, namely, gametocyte-enriched blood. As the reviewer points out, timing is important, and samples from the period when TFs are maximally expressed are optimal for ChIP-seq. However, when parasites in the asexual stages are present, the background becomes higher. Thus we usually use gametocyte-enriched blood for ChIP-seq when expression of the TF is observed in mature gametocytes. The exception was our ChIP-seq analysis of AP2-G, because is not present in mature gametocytes.

- *Fig 4c is an example of great overlap of peaks, but it would be helpful if the authors could quantify the overlaps between experiments (and describe the overlap parameters used).*

Response: According to the comment, we have created a Venn diagram of overlapping peaks (attached below). However, the peaks used for this Venn diagram were selected after peakcalling via fold-enrichment values. Thus, even if the counterpart of a peak is absent in these selected peaks (non-overlapping peaks in the Venn diagram), it does not indicate that it is absent in the original read map. We believe the overlap of peaks would be estimated more correctly in the heat maps.

Author response image 1.

Legged: The Venn diagram shows the number of common peaks between these ChIP seq experiments (distance of peak summits < 150



- *Additionally, how were the promoter coordinates used for each gene when they associate ChIP peaks to a gene target. Did the authors choose 1-2kb? Or use a TSS/5utr dataset such as Adjalley 2016 or Chappell 2020?*

Response: We selected a 1.2 Kbp region for target prediction based on our previous studies. As the reviewer pointed out, target prediction using TSS information may be more accurate. However, reliable TSS information is not available for *P. berghei* to the best of our knowledge.

The two papers are studies on *P. falciparum*.

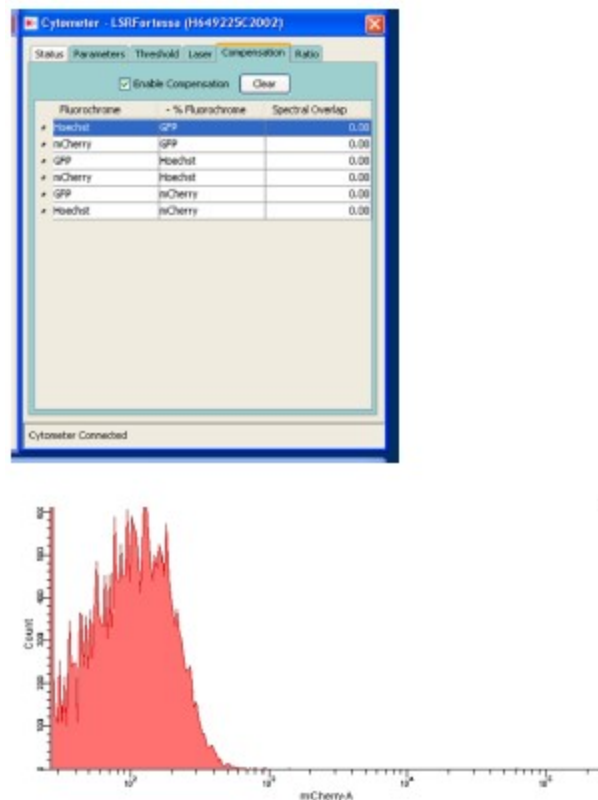
- In the absence of evidence of physical interaction, it remains unclear if AP2-FG and PFG actually interact directly or as part of the same complex. A more detailed characterisation with IPs/co-IPs followed by mass spectrometry of the GFP-tagged version of PFG in the presence and absence of AP2-FG would be highly informative.*

Response: Thank you for the comment. Even when these two TFs occupy the same genomic region, it cannot be conclusively said that they exist at the same time in the region: they might competitively occupy the region. However, we showed that the cAP2-FG peaks disappear from the region when PFG was disrupted, while sAP2-FG peaks remain. We believe that this is evidence that the two TFs physically interact with each other.

- It was not clear if the assessment of motif binding using cytometry was performed using all the required controls and compensation. This section should be clarified.*

Response: Thank you for the comment. Condensation was performed using parasites expressing a single fluorescent protein. The results are attached below. The histogram of mCherry using control parasites expressing GFP under the control of the HSP70 promoter is also attached.

Author response image 2.



However, we found that descriptions of the filters for detecting red signals were not correct. This assay was performed using parasites which expressed GFP constitutively and mCherry under the control of the p28 promoter. These two fluorescent proteins were excited by independent lasers (488 and 561, respectively), and the emission spectra were detected using independent detectors (through 530/30 and 610/20 filters, respectively). We have revised the description regarding our FACS protocols as follows:

“Flow cytometric analysis was performed using an LSR-II flow cytometer (BD Biosciences). In experiments using 820 parasites, the tail blood from infected mice was selected via gating with forward scatter and staining with Hoechst 33342 (excitation = 355 nm, emission = 450/50). The gated population was then analyzed for GFP fluorescence (excitation = 488 nm, emission = 530/30) and RFP fluorescence (excitation = 561 nm, emission = 610/20). In the promoter assay (using parasites transfected with a centromere plasmid), the tail blood from infected mice was selected via gating with forward scatter and staining with Hoechst 33342 (excitation = 355 nm, emission = 450/50), followed by GFP fluorescence (excitation = 488 nm, emission = 530/30). The gated population was analyzed for mCherry fluorescence (excitation = 561 nm, emission = 610/20). Analysis was performed using the DIVER program (BD Biosciences).”

Minor points:

- Page 4, line 37: The authors should specify the timing of expression of AP2-FG on the text.

Response: We have added the following description to the text.

“The timing of the expression was approximately four hours later than that of AP2-FG, which started at 16 hpi (9).”

- *Ref 9 and 17 are repeated*

Response: Thank you for pointing this out. We have corrected this.

- *Fig 1D and 1F do not have scale bars*

Response: We have added scale bars to Fig. 1D.

We have not changed Fig. 1F, because we believe that the scales can be estimated from the size of the erythrocyte.

- *Page 5, line 29-30. Could the authors specify how many and which of the de-regulated genes have a PFG in their promoter.*

Response: Thank you for the comment, As described in a later section (page 7; Impact of PFG disruption on the expression of AP2-FG target genes), among the 279 genes significantly downregulated in PFG(-) parasites, 165 genes were targets for PFG (unique for PFG or common for sAP2-FG and PFG). In contrast, only four genes were targets unique to sAP2-FG. Therefore, 165 genes harbor the upstream peaks of PFG. These genes are shown in Table S1.

- *Fig 5F. in the methods associated with this figure there seems to be a mixup with the description of the lasers. In addition, given the spillover of the red and green signal between detectors this experiment needs compensation parameters. The authors should provide the gating strategy before and after compensation as this is critical for the correct calculation of the number of red parasites. Indeed, the lowest red cloud on the gate shown could be green signal spill over.*

Response: Thank you for the comment. As described above, there were some incorrect descriptions about the conditions of our FACS protocols in the methods section. We have revised them.

- *-Page 7, line 19. Could the authors explicitly say in the text that the 810 genes are those with 1 (or more?) PFG peaks in their promoter (out of a total of 1029) to best guide the reader. Additionally, it is important to define the maximum distance allowed between a peak and CDS for it to be associated with said CDS.*

Response: We have revised Table S2 by adding the nearest genes. The revised table shows the relationship between a PFG peak and its nearest genes, together with their distances.

- *Page 7, line 45: fig 6c, not 5c*

Response: Thank you for the comment. We have corrected this.

- *Page 7 last paragraph: This section is very hard to follow. For instance, on line 50 do the authors mean that the sAP2-FG unique targets are LESS de-regulated? On line 51: do the authors mean unique targets of cAP2-FG or unique targets of PFG? Line 53: do the authors mean that genes expressed in the "common" category are LESS de-regulated than the PFG unique targets?*

Response: We are sorry for the lack of clarity; after reviewing the manuscript, it appears to be unclear what the fold change means in this section. Here, fold change means the ratio of PFG(-)/wild type. Thus “High log₂(fold change) value” means that the genes were less downregulated. We have revised the description as follows:

“The log₂ distribution (fold change = PFG(-)/wild type) in the three groups of target genes showed that the average value was significantly higher (i.e., less down-regulated) in targets unique to sAP2-FG than in the other two groups (targets unique to cAP2-FG or common targets for both), with p-values of 1.3×10^{-10} and 1.4×10^{-5} , respectively, by two-tailed Student’s t-test (Fig. 6F). In addition, the average log₂ (fold change) value of the common target genes was relatively higher (i.e., less down-regulated) than that of targets unique to PFG, suggesting that transcriptional activation by sAP2-FG partly complements the impact of PFG disruption on these common targets.”

- Page 8, line 42: Fig 6G, not 5G

Response: Thank you for pointing this out. We have corrected this.

Reviewer #3 (Recommendations For The Authors):

1. The gene at the center of this study (PBANKA_0902300) was identified in an earlier genetic screen by Russell et al. as being a female specific gene with essential role in transmission and named Fd2 (for female-defective 2). Since this name entered the literature first and is equally descriptive, the Fd2 name should be used instead of PFG to maintain clarity and avoid unnecessary confusion. Surprisingly, this study is neither cited nor acknowledged despite a preprint having been available since August of 2021. This should be remedied.

Response: Thank you for the comment. We have added the paper by Russell et al. accordingly and mentioned the name FD2 in the revised manuscript. However, we have retained the use of PFG throughout the paper. We believe that this usage of PFG shouldn’t be confusing, as FD2 has only been used in one previous paper. We have added the following:

“Through this screening, a gene encoding a 2709 amino acid protein with two regions highly conserved among Plasmodium was identified (PBANKA0902300, designated as a partner of AP2-FG (PFG; Fig. 1A). This gene is one of the P. berghei genes that were previously identified as genes involved in female gametocyte development (named FD2), based on mass screening combined with single cell RNA-seq (ref).”

1. While it isn't really important how the authors came to arrive at studying the function of Fd2, the rationale/approach given in the first paragraph of the result section seems far too broad to lead to Fd2, given that it lacks identifiable domains and many other ortholog sets exist across these species.

Response: We selected this gene from the list of AP2-G targets as a candidate for a sequence-specific TF based on the hypothesis that the amino acid sequences of DNA-binding domains are highly conserved. We successfully identified two TFs (including PFG) using this method. However, there may be TFs that do not fit this hypothesis which are also targets of AP2-G. In fact, we were unable to identify HDP1 as a TF candidate, despite being a AP2-G target.

1. Fig. 1A-C: Gene IDs for the orthologs should be provided, as well as the methodology for generating the alignments.

Response; We have added the gene IDs and method for alignment in the legend as follows:

(A) Schematic diagram of PFG from *P. berghei* and its homologs in apicomplexan parasites. Regions homologous to Regions 1 and 2, which are highly conserved among *Plasmodium* species, are shown as yellow and blue rectangles, respectively. Nuclear localization signals were predicted using the cNLS mapper (http://nls-10.mapper.iab.keio.ac.jp/cgi-bin/NLS_Mapper_form.cgi). The gene IDs of *P. berghei* PFG, *P. falciparum* PFG, and their homologs in *Toxoplasma gondii*, *Eimeria tenella* and *Vitrella brassicaformis* are PBANKA_0902300, PF3D7_1146800, TGGT1_239670, ETH2_1252400, and Vbra_10234, respectively.

(C) The amino acid sequences of Regions 1 and 2 from *P. berghei* PFG and its homologs from other apicomplexan parasites in (A) were aligned using the ClustalW program in MEGA X. The positions at which all these sequences have identical amino acids are indicated by two asterisks, and positions with amino acid residues possessing the same properties are indicated by one asterisk.

1. Figure 2: The Phenotype of Fd2 knockout should be characterized more comprehensively.

It remains unclear whether Δ Fd2 parasite generate the same number of females but these are defective upon fertilization or whether there is also a decrease in the number of female gametocytes. Is the defect just post-fertilization and zygotes lyse or are there fewer fertilization events? If so is activation of female GCs effected?

The number of male and female gametocytes should be quantified using sex-specific markers not affected by Fd2 knockout rather than providing a single image of each. The ability of Δ Fd2 GCs should also be evaluated.

This is also important for the interpretation of Fig 2G. Is the down-regulation of the genes due to fewer female GCs or are the down-regulated genes only a subset of female-specific genes.

Response: In PFG(-) parasites, the rate of conversion into zygotes of female gametocytes decreased, and zygotes had lost capacity for developing into ookinetes. This indicates that gametocyte development (i.e., the ability to egress the erythrocyte and to fertilize) and zygote development were both impaired. This phenotype is consistent with the observation that genes expressed in female gametocytes are broadly downregulated. PFG is a TF, and its disruption led to decreased expression of hundreds of female genes. Thus, the observed phenotype may be derived from combined decreased expression of these genes. We believe further detailed phenotypic analyses will not generate much novel information on this TF. Instead, RNA-seq data in PFG(-) parasites and the targetome have promise in helping to characterize the functions of this TF.

1. Figure 3: what fraction of down-regulated genes have the Fd2 10mer motif?

Response: Thank you for the question. We investigated the upstream binding motifs of these genes. Of the 279 significantly down-regulated genes (containing 165 targets), 161 genes harbor the motif (including nine-base motifs that lack one lateral base which is likely not essential for binding) in their upstream regions (within 1,200 bp from the first methionine codon). However, this result has not been described in the revised manuscript because it is more important whether these regions harbor PFG peaks (upstream motifs can exist without being involved in the binding of PFG).

1. *sAP2-FG (single) vs cAP2-FG (complex) nomenclature is confusing and possibly misleading since few TFs function in isolation and sAP2-FG likely functions in a complex that doesn't contain Fd2, possibly with another DNA binding protein that binds the TGCACA hexamer. The name for the distinct peaks should refer to the presence or absence of Fd2 in the complex, or maybe simply refer to them as complex A & B.*

Response: As shown in the DIP-seq analysis results, AP2-FG can bind the motif by itself. In contrast, AP2-FG must form a complex with PFG to bind to the ten-base motif. The complex and single forms are named according to this difference (the presence or absence of PFG) and used solely in its relation with PFG. We wrote “In the following, we refer to the form with PFG as cAP2-FG or the complex form, and the form without PFG as sAP2-FG or the single form.” We believe that the nomenclature has sufficient clarity. However, we have partially (underlined) revised certain sentences in the discussion section as follows.

“As the expression of PFG increases via this mechanism, AP2-FG recruited by PFG (cAP2FG) increases and eventually becomes predominant in the transcriptional regulation of female gametocytes.”

“This suggests that the promoter of the CCP2 gene, which is a target of PFG only, is still active in AP2-FG(-)820 parasites.”

We recently reported that the TGCACA motif is a cis-activation motif in early gametocytes and important for both male and female gametocyte development. Thus we speculate that sAP2-FG is not involved in cis-activation by the TGCACA motif. The p-value of the six-base motif is indeed comparable to that of the five-base motif. However, the pvalue (calculated by Fisher's exact test) in six-base motifs tend to be lower than that calculated in five-base motifs, because the population is much large. We speculate that there is a sequence-specific TF that may be expressed in early gametocytes and bind this motif, independently of AP2-FG.

1. *I compared the overlap of peaks in the 4 ChIP-seq data sets:*

90% of the Fd2 peaks are shared with AP2-FG (binding 24% of shared peaks is lost in Δ AP2FG)

10% are bound by Fd2 alone (binding at 35% of Fd2 is lost in Δ AP2-FG)

75% of Fd2 peaks are bound independently of AP2-FG

47% of AP2-FG peaks shared with Fd2 (binding at 71% of shared peaks is lost in Δ Fd2)

53% of AP2-FG peaks are bound only by AP2-FG (but binding at 82% of AP2-FG only peaks is still lost in the Δ Fd2)

Binding at 78% of all AP2-FG peaks is lost in Δ Fd2

This indicates that much of AP2-FG binding in regions even in regions devoid of Fd2 still depends on Fd2. What are possible explanations for this?

https://elifep-rp.msubmit.net/eliferp_files/2023/04/03/00117573/00/117573_0_attach_10_17936_convrt.pdf

Response: In the ChIP-seq of AP2-FG in the absence of PFG, 441 peaks are still called. This means that at least 441 binding sites for AP2-FG independent of PFG exist. This is a straightforward conclusion from our ChIP-seq data. On the other hand, simple deduction of peaks between two ChIP-seq experiments (AP2-FG peaks minus PFG peaks) is not a precise method for determining sAP2-FG. Peak-calling is independently performed in each ChIP-seq

experiment. Thus, peaks remaining after the deduction between two experiments can still contain peaks that are actually common, but which are differentially picked up through the process of peak calling. Even when using data obtained by the same ChIP-seq experiment, markedly different numbers of peaks are called according to the conditions for peak calling (in contrast, common peaks between two independent experiments increase the reliability of the data). If wanting to identify sAP2-FG peaks via comparisons between AP2-FG peaks and PFG peaks, the reviewer has to increase the number of PFG peaks by reducing the peak-calling threshold until the number of overlapping peaks between AP2-FG and PFG are saturated, and then deduce the overlapping peaks from the AP2-FG peaks. However, as described above, for the purposes of estimating the number of sAP2-FG, it would be better to perform ChIP-seq of AP2-FG in the absence of PFG.

1. *Possible explanations of why recombinant Fd2 doesn't bind the TGCACA hexamer. It would also be good to note that the GCTCA AP2-FG motif found in Fig4G is now perfect match for the motif identified by protein binding microarray in Campbell et al.*

Response: It is not known what sequence recombinant PFG binds. The TGCACA motif is not enriched in PFG peaks. If the reviewer is referring to AP2-FG, our findings that the recombinant AP2 domain binds the five-base motif strongly suggests that other TFs recognize this motif. As described in our response to comment 9, we recently reported that TGCACA is a cis-activating sequence important for the normal development of both male and female gametocytes. Therefore, we currently speculate that this motif is a binding motif of other TFs and is independent of AP2-FG.

We have mentioned the protein binding microarray data in the Results section as follows.

“The most enriched motif matched well with the binding sequence of the AP2 domain of *P. falciparum* AP2-FG, which was reported by Campbell et al.”

1. *What might explain the strong enrichment for TGCACA in ChIPseq but when pulled down by AP2-FG DBD: another binding partner? requires more of AP2-DF than just DBD?*

Response: As described above in our response to comment 6, we have recently submitted a preprint studying the roles of the remodeler subunit PbARID in gametocyte development. We reported that the remodeler subunit is recruited to the six-base motif and that the motif is a novel cis-activation element for early gametocyte development. We speculate that a proportion of AP2-FG targets are also targets of a TF that recognizes this motif and recruits the remodeler subunit. These two TFs may be involved in the regulation of early gametocyte genes but function independently.

1. *Calling DNA pulldown with recombinant AP2-FG DNA-binding domain DNAImmunoprecipitation sequencing (DIP-seq) is confusing since there are no antibodies involved. Describing it directly as a pulldown of fragmented DNA will be clearer to the reader.*

Response: Thank you for the comment. We have also recognized this discrepancy. However we called the method DIP-seq because the original paper reporting this method used this name, wherein it did not use antibodies to capture the MBP-fusion recombinant protein. Our experiment was performed using essentially the same methods, and thus we retained the name.

1. *The legends and methods are very sparse and should include substantially more detail.*

Response: Thank you for the comment. We have revised the description of the FACS experimental method for clarity.

1. *BigWig files for all ChIPseq enrichment used for analysis in this study need to be provided.*

(two replicates each of: Fd2 in WT, Fd2 in Δ AP2-GF, AP2-FG in WT, AP2-FG in Δ Fd2)

Response: We have deposited the BigWig files to GEO (GSE.226028 and GSE114096).

1. *Tables of ChIP data need to have both summits and peaks and need to list nearest gene. Also the ChIPseq peaks for Fd2 are surprisingly broad (ChIP peaks are very large, e.g. 68% of Fd2 peaks (dataset2) are greater than 1000kb) give its specificity for a long motif. Why is this?*

Response: We have revised Table S2 to include the nearest genes. We are unsure why peaks in the over 1000-bp peak region exist in such high proportions. However, this proportion was also high in our previous ChIP-seq data. Therefore, we speculate that this is a tendency of peak-calling by MACS2. We did not use these values in this paper. For example, targets were predicted using peak summits, and binding motifs were calculated using the 100-base regions around peak summits.

1. *Figure 5E: The positions of the 10mer and 5mer motifs in the promoter should be indicated as well as the length of the promoter. Moreover, mutation of just the 5bp motifs would be valuable to understand if 10mer is sufficient for expression of the reporter.*

Response: Thank you for the comment. We have revised the figure accordingly. The majority of female-specific promoters only harbor ten-base motifs. Thus the ten-base motif is sufficient for evaluating reporter activity (i.e., it would function without five-base motifs).

1. *How is AP2-FG expression affected in Δ Fd2 and vice versa?*

Response: According to our previous microarray data, PFG expression was not significantly downregulated by disruption of AP2-FG. This may be because PFG transcriptionally activates itself through a positive feedback loop after being induced by AP2-G. Similarly, according to our present study, AP2-FG expression was not downregulated by PFG disruption. This may be because AP2-FG is transcriptionally activated by AP2-G.

1. *The single cell data in Russell et al. could easily be used to indicate the order of expression.*

Response: Determining the expression order of gametocyte TFs via the single cell RNA-seq data from Russell et al. is difficult, because only a small number of parasite cells were considered to be in the early gametocyte stage in this study. This is because the parasites were cultured for 24h before the analysis. The analysis suggested by the reviewer may be possible via single cell RNA-seq, but the experiments must be performed with more focus on the early gametocyte stage.

1. *A discussion of the implication of *P. falciparum* transmission would be appreciated.*

Response: Thank you for the comment. We have added the following to the Discussion section:

"*P. falciparum* gametocytes require 9-12 days to mature, which is much longer than that of *P. berghei*. Meanwhile, it has been reported that the ten-base motif is highly enriched in the upstream regions of female-specific genes also in *P. falciparum*. Thus, despite the difference in maturation periods, PFG is likely to play an important role in the transcriptional regulation of female *P. falciparum* gametocyte development."

- 1. The lack of identifiable DNA binding domains in Fd2 is intriguing given the strong sequence-specificity. Do the authors think they have identified a new DNA-binding fold ? Alphafold of the orthologs with contiguous regions 1&2 might offer insight.*

Response: We speculate that these regions function as DNA binding domains. We performed analysis using Alfafold2 according to the comment. However, the predicted structure of the region was not similar to any other canonical DNA-binding domains. Thus, it may be a novel DNA-binding fold as the reviewer mentioned. Further studies such as binding assays using recombinant proteins would be necessary to confirm this, but thus far we have not successfully obtained the soluble proteins of these regions.